

- Maximum Likelihood Estimation (MLE)** : The MLE method maximises the likelihood function to obtain an estimate of θ as a value $\hat{\theta}$ such that $L(\hat{\theta}) \geq L(\theta)$ for all θ .
- Method of Moments** : The method first equates the moments of the theoretical distribution (expressed in terms of parameters) with their corresponding sample moments. It then proceeds to solve the equations to obtain the estimates of the population parameters.

2.8 SOME USEFUL BOOKS

Bhaumik, Sankar (2017), *Principles of Econometrics: A Modern Approach Using E-views*, Oxford University Press.

Rasch Dieter, Schott Dieter (2018), *Mathematical Statistics*, Wiley.

2.9 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) A parameter relates to the unknown in the population (e.g. mean, variance). A statistic relates to the estimated mean or variance on the basis of a sample drawn from a population. A statistic is a function of sample observations.
- 2) A point estimate of a parameter is a specific value of a 'statistic'. Any particular estimate obtained from a sample is a 'point estimate'. If T_1 and T_2 are two estimates such that ' $P(T_1 \leq \theta \leq T_2) = 1 - \alpha$ ' (where α is the level of significance), the interval (T_1, T_2) is termed as an 'interval estimate'. It is also called a 'confidence interval'.
- 3) While the estimator will be the same for multiple samples, the estimate will vary from sample to sample. The distribution of many such point estimates, obtained from different samples, is called the 'sampling distribution' of the 'statistic'.
- 4) (i) unbiasedness, ii) consistency, iii) efficiency and iv) sufficiency.
- 5) (i) $\lim_{n \rightarrow \infty} [E(T) - \theta] = 0$ and (ii) $\lim_{n \rightarrow \infty} Var(T) = 0$.
- 6) If A_n is the sample mean for any n i.i.d. random variables with mean μ , then for the WLLN, then any positive number α , we will have $\lim_{n \rightarrow \infty} P(|A_n - \mu| < \alpha) = 1$. For the SLLN, we will have $\lim_{n \rightarrow \infty} P(|A_n - \mu| = 0) = 1$.

Check Your Progress 2

- 1) A hypothesis is said to be complete if all the parameters of the population from which the sample (to test the hypothesis) is drawn are completely specified. Such hypotheses are also called 'simple hypotheses'. This is in

contrast to a 'composite hypothesis', which refers to a hypothesis in which one or more of the parameters are either not exactly specified or not specified at all.

- 2) If the alternative hypothesis is specified as $H_1: \mu \neq \mu_0$ then the test is called the two-tailed test. If the alternative hypothesis is one-directional like $H_1: \mu < \mu_0$ or $\mu > \mu_0$ then the test is called a one-tailed test. It is either a left tailed test or a right tailed test depending on whether the inequality is $<$ or $>$ in H_1 .
- 3) Rejecting a null hypothesis, when it is actually true, is termed as type-I error. The opposite, i.e., 'accepting an alternative hypothesis when it is actually false' is termed as type-II error. Between the two, the type-I error is considered more grave. Hence, we try to ensure that this is kept as low as possible. This level at which we keep the test is called the 'level of significance', usually denoted by α , and this is kept at either 1 percent or 5 percent level. This is the maximum permissible level of committing the type-I error.
- 4) It states that the best values of the unknown are those for which the 'sum of the squares of the residuals', $S = \sum_{i=1}^m e_i^2$, is the least. The method of least squares gives a unique solution when the number of the normal equations is exactly equal to the number of unknown parameters or the coefficients of the regression model.
- 5) The properties of a maximum likelihood estimator are as follows: (a) It is consistent, efficient and sufficient, (b) it is not necessarily unbiased but can be made unbiased, (c) it tends to be distributed normally for large samples, and (d) it is invariant to functional transformation.

UNIT 3 REVIEW OF MATRIX ALGEBRA*

Structure

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3.0 OBJECTIVES

After going through this unit, you will be able to

- carry out various matrix operations such as multiplication and inverse;
- find out the determinant and rank of a matrix;

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- define eigenvalue and eigenvector and compute them;
- identify some special matrices which are relevant to econometrics;
- compute Kronecker product and vec-operator; and
- calculate Jacobian matrix, Gradient matrix and Hessian matrix.

3.1 INTRODUCTION

Matrix algebra is an integral part of econometrics. It not only helps in presentation of multiple variables in a concise form, but also facilitates various mathematical operations. Further, matrix algebra allows us to generalise results from a small number of variables to larger ones.

You have already studied matrix algebra in the course MEC 203: Quantitative Methods. In the present Unit we provide an overview of certain concepts which are usually applied in econometrics. Our emphasis is on certain special matrices and their properties. Although the coverage of the topics in the present Unit is limited, we have attempted to provide examples of the concepts for easier comprehension.

3.1 BASIC NOTATIONS

Let us begin with the definition of a matrix. As you know, matrix is an array of numbers. A matrix is denoted by an upper case alphabet in boldface (e.g., **A**) and its $(i, j)^{\text{th}}$ element (the element at the i^{th} row and j^{th} column) is denoted by corresponding lower case alphabet with subscript i, j (e.g., a_{ij}). The dimension or order of a matrix is given by $m \times n$, when the matrix has m rows and n columns. For example, matrix **A** can be expressed as

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

If $m = n$ then the matrix is called a 'square matrix'. An $n \times 1$ matrix is an n -dimensional column vector. Similarly, an $1 \times n$ matrix is a row vector. Every vector will be denoted by a lower-case alphabet in boldface (e.g., **a**) and its element is denoted by corresponding lower case alphabet with subscript i (e.g., a_i). An 1×1 matrix is just a scalar. For a matrix **A** its i^{th} column is denoted by $\mathbf{a}_i$.

Alternatively, $\mathbf{A} = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n]$

where $\mathbf{a}_i = [a_{1i} \quad a_{2i} \quad \cdots \quad a_{mi}]^T \quad \forall i = 1, \dots, n$

A matrix is said to be diagonal if its off-diagonal elements (i.e., $a_{ij}, i \neq j$) are all zero and at least one of its diagonal elements is non-zero, i.e., $a_{ii} \neq 0$ for some $i = 1, 2, \dots, n$. A diagonal matrix whose diagonal elements are all one is an identity matrix denoted as **I**; we also write the $n \times n$ identity matrix as $\mathbf{I}_n$.

A matrix **A** is said to be 'lower-triangular' if $a_{ij} = 0$ for $i < j$. Similarly, a matrix is said to be 'upper-triangular' if $a_{ij} = 0$ for $i > j$. We denote **0** (null matrix) as a matrix whose all elements are zero. The transpose of a $m \times n$ matrix

$A = (a_{ij})$ is $n \times m$ matrix. Thus, $A' = (a_{ji})$. Note that the transpose of a matrix can be denoted by the super-script “T”. Thus, for the matrix A , the notations A' and A^T are used interchangeably.

3.3 MULTIPLICATION OF MATRICES

If A is $m \times n$ matrix and B is $p \times q$ matrix then the product of A and B is defined as

$$AB = (C_{ij})_{m \times q} \text{ where } C_{ij} = (\sum_{j=1}^n a_{ij}b_{jk}) \text{ provided } n = p.$$

The dimension of AB is given by $m \times q$.

When the order of A and B are such that AB is defined, we say that A and B are conformable. For a real-valued square matrix A , we say that A is ‘orthogonal’ if $A'A = AA' = I$... (3.1)

For any two matrices A and B for which AB is defined,

$$(AB)^T = B^T A^T \text{ ... (3.2)}$$

If we multiply a square matrix A by itself, that is AA , we obtain A^2 .

Let us consider an example of matrix multiplication.

Example 3.1:

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \quad B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$$

$$\text{Hence, } AB = \begin{pmatrix} a_{11} \cdot b_{11} + a_{12} \cdot b_{21} & a_{11} \cdot b_{12} + a_{12} \cdot b_{22} \\ a_{21} \cdot b_{11} + a_{22} \cdot b_{21} & a_{21} \cdot b_{12} + a_{22} \cdot b_{22} \end{pmatrix}$$

Let us consider one more example.

Example 3.2:

$$\text{If } A = \begin{pmatrix} 1 & 2 & 0 \\ 4 & 5 & 6 \end{pmatrix} \quad B = \begin{pmatrix} 4 & 0 \\ 5 & 2 \\ 6 & 1 \end{pmatrix}. \text{ Find } AB.$$

$$AB = \begin{pmatrix} 1 \cdot 4 + 2 \cdot 5 + 0 \cdot 6 & 1 \cdot 0 + 2 \cdot 2 + 0 \cdot 1 \\ 4 \cdot 4 + 5 \cdot 5 + 6 \cdot 6 & 4 \cdot 0 + 5 \cdot 2 + 6 \cdot 1 \end{pmatrix} = \begin{pmatrix} 14 & 4 \\ 77 & 16 \end{pmatrix}$$

If $AB = BA$, then we say that both matrices commute.

Any matrix B satisfying $B^{k-1} \neq 0$ and $B^k = 0$ is said to be ‘nilpotent matrix’ of index k . The matrix $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ is nilpotent of index 2.

The following are a few theorems related to matrix multiplication (we do not provide the proofs here; you can refer to the course MEC 203).

- (i) $(A + B)^2 = (B + A)^2$
- (ii) $(A + B)^2 = A^2 + AB + BA + B^2$
(because, in general, $AB \neq BA$)
- (iii) $(A + B)^2 = (A + B)(A + B)$
- (iv) $A'A = 0$ iff $A = 0$

- (v) $\mathbf{AB} = \mathbf{0}$ iff $\mathbf{A}'\mathbf{AB} = \mathbf{0}$
- (vi) $\mathbf{AB} = \mathbf{AC}$ iff $\mathbf{A}'\mathbf{AB} = \mathbf{A}'\mathbf{AC}$
- (vii) $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$ for conformable matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$
- (viii) $\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$ for conformable matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$
- (ix) $(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'$
- (x) $(\mathbf{ABC})' = \mathbf{C}'\mathbf{B}'\mathbf{A}'$

3.4 DETERMINANT AND TRACE OF A MATRIX

Given a square matrix $\mathbf{A}$, let A_{ij} denote the sub-matrix obtained from $\mathbf{A}$ by deleting its i^{th} row and j^{th} column. The determinant of $\mathbf{A}$ is denoted by the symbol $\det(\mathbf{A})$. In some texts, you can see that determinant is denoted by the symbol Δ or $|\mathbf{A}|$.

We can find out the determinant of a matrix $\mathbf{A}$ by the following formula:

$$\det(\mathbf{A}) = \sum_{i=1}^m (-1)^{i+j} a_{ij} \det(A_{ij}) \quad \dots (3.3)$$

for any $j = 1, \dots, n$ where $(-1)^{i+j}$. Note that $\det(A_{ij})$ is called the *cofactor* of a_{ij} . This definition is based on the cofactor expansion along the j^{th} column. Equivalently, we can define the determinant by using the cofactor expansion along the i^{th} row.

$$\det(\mathbf{A}) = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det(A_{ij}) \text{ for any } i = 1 \dots n. \quad \dots (3.4)$$

The determinant of a scalar is the scalar itself. The determinant of a 2×2 matrix $\mathbf{A}$ is given by

$$|\mathbf{A}| = (a_{11}a_{22} - a_{12}a_{21})$$

You can find out the determinant of a higher order matrix by using its *cofactors*.

Example 3.3:

$$\mathbf{A} = \begin{bmatrix} 2 & 4 \\ 1 & 5 \end{bmatrix}. \text{ Find } |\mathbf{A}|.$$

$$|\mathbf{A}| = 2 \times 5 - 1 \times 4 = 6.$$

A square matrix is singular if its determinant is equal to zero. It is called singular because its inverse cannot be calculated. A square matrix with non-zero determinant is said to be 'non-singular' (implies, the matrix has an 'inverse' or the matrix is 'invertible').

Clearly, $\det(\mathbf{A}) = \det(\mathbf{A}')$

For a scalar c and an $n \times n$ matrix $\mathbf{A}$

$$\det(c\mathbf{A}) = c^n \det(\mathbf{A})$$

You should note that the determinant of a square matrix with a column (or row) of zeros, is zero. Also, the determinant of a diagonal or triangular matrix is simply the product of all diagonal elements. It can also be shown that the

determinant of the product of two square matrices of the same size is the product of their determinants.

$$\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B}) = \det(\mathbf{BA}) \quad \dots (3.5)$$

Also, for an $m \times m$ matrix $\mathbf{A}$ and $n \times n$ matrix $\mathbf{B}$

$$\det(\mathbf{A} \otimes \mathbf{B}) = \det(\mathbf{A})^m \det(\mathbf{B})^n \quad \dots (3.6)$$

The symbol $\otimes$ is the Kronecker product, which we will discuss later in Section 3.10.

If $\mathbf{A}$ is orthogonal matrix, we know that $\mathbf{AA}' = \mathbf{I}$.

Therefore,

$$\det(\mathbf{I}) = \det(\mathbf{A}'\mathbf{A}) = [\det(\mathbf{A})]^2 \quad \dots (3.7)$$

As the determinant of an identity matrix is one, the determinant of an orthogonal matrix must be either 1 or -1 .

The *trace* of a square matrix is the sum of its diagonal elements, i.e.,

$$\text{trace}(\mathbf{A}) = \sum_i a_{ii} = \text{tr}(\mathbf{A}) \quad \dots (3.8)$$

For example, $\text{trace}(\mathbf{I}_n) = n$

Let $\mathbf{A}$ and $\mathbf{B}$ be square matrices of the same order. Let λ and μ be scalars. Then

- (i) $\text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr}(\mathbf{A}) + \text{tr}(\mathbf{B})$
- (ii) $\text{tr}(\lambda\mathbf{A} + \mu\mathbf{B}) = \lambda\text{tr}(\mathbf{A}) + \mu\text{tr}(\mathbf{B})$
- (iii) $\text{tr}(\mathbf{A}) = \text{tr}(\mathbf{A}')$
- (iv) $\text{tr}(\mathbf{AA}') = \text{tr}(\mathbf{A}'\mathbf{A}) = \sum_{ij} a_{ij}^2$
- (v) $\text{tr}(aa') = \text{tr}(a'a) = \sum_i a_i^2$ for any vector a .

3.5 INVERSE OF A MATRIX

A non-singular matrix $\mathbf{A}$ possesses a unique inverse $\mathbf{A}^{-1}$ in the sense that $\mathbf{AA}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$. Note that a singular matrix cannot be inverted.

Given an invertible matrix $\mathbf{A}$, its inverse can be calculated as $\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \mathbf{F}'$.

Here, $\mathbf{F}$ is the matrix of the cofactors of $\mathbf{A}$. Recall that the $(i,j)^{\text{th}}$ element of $\mathbf{F}$ is the cofactor $(-1)^{i+j} \det(\mathbf{A}_{ij})$. The matrix $\mathbf{F}'$ is known as the *adjoint* of $\mathbf{A}$.

Example 3.4:

Find the inverse of matrix $\mathbf{A}$.

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

First, find the determinant of $\mathbf{A}$, which is $(a_{11}a_{22} - a_{12}a_{21})$

Second, find the matrix of cofactors, which is $\begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$.

The inverse of matrix $\mathbf{A}$ is

$$\mathbf{A}^{-1} = \frac{1}{a_{11}a_{22} - a_{12}a_{21}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$$

Properties of the Inverse of a Matrix

For any non-singular matrix $\mathbf{A}$

- (i) $(\lambda \mathbf{A})^{-1} = \frac{1}{\lambda} \mathbf{A}^{-1}$ (for any scalar quantity $\lambda \neq 0$)
- (ii) $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$
- (iii) $(\mathbf{A}^{-1})^T = (\mathbf{A}^T)^{-1}$
- (iv) $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$ if $\mathbf{B}$ is non-singular and of the same order as $\mathbf{A}$
- (v) If $\mathbf{A}$ and $\mathbf{B}$ commute and $\mathbf{B}$ is non-singular then $\mathbf{A}$ and $\mathbf{B}^{-1}$ commute.
- (vi) For a diagonal matrix $\mathbf{A}$, note that $\mathbf{A}^{-1}$ is also diagonal with the diagonal elements a_{ii}^{-1} .

3.6 RANK OF A MATRIX

Before proceeding to define the rank of a matrix, we need to define 'row-reduced echelon form of a matrix'. A matrix is said to be in *row echelon form*, if it satisfies the following:

- (1) all rows having only zeros at bottom of $\mathbf{A}$
- (2) all the non-zero rows of $\mathbf{A}$ are, as the row number increases then the number of zeros at the beginning of the rows also increases.
- (3) In order to convert a matrix into row echelon form we can perform two operations on the rows, i.e., scalar multiplication and addition/subtraction. However, it is important to mention that we have to perform the operation on all the elements of a row.

Now let us define the rank of a matrix. The rank of matrix $\mathbf{A}$ is the maximum number of linearly independent rows in $\mathbf{A}$. Rank can also be defined as the maximum number of linearly independent columns in a matrix.

Hence, we can say rank of $\mathbf{A} = \text{rank of } (\mathbf{A}^T)$

The rank of $\mathbf{A}$ is the total number of non-zero rows in the row echelon form of $\mathbf{A}$.

Example 3.5:

Find the rank of matrix $\mathbf{A}$.

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 1 & -2 \\ 2 & 1 & 3 & -1 \\ 2 & 0 & 1 & 4 \end{pmatrix}$$

Reduce this matrix to row reduce echelon form by replacing R_2 of matrix by (row 2 – 2 times row 1)

$$\begin{pmatrix} 1 & 2 & 1 & -2 \\ 0 & -3 & 1 & 3 \\ 2 & 0 & 1 & 4 \end{pmatrix}$$

By replacing row 3 of above matrix by (row 3 – 2 times row 1)

$$\begin{pmatrix} 1 & 2 & 1 & -2 \\ 0 & -3 & 1 & 3 \\ 0 & -2 & -1 & 8 \end{pmatrix}$$

By replacing row 3 of above matrix by (3 times row 3 – 2 times row 2) we get

$$\begin{pmatrix} 1 & 2 & 1 & -2 \\ 0 & -3 & 1 & 3 \\ 0 & 0 & -5 & 18 \end{pmatrix}$$

This matrix is in echelon form and there is no zero-row. The rows of the matrix are linearly independent. Thus, the rank of the matrix is three.

You should note the following two points: (i) If the matrix **A** is of dimension $m \times n$ where $m \neq n$, then rank of the matrix is given by: $\text{rank}(\mathbf{A}) \leq \min\{m, n\}$. (ii) For a square matrix to be invertible, it must be of full rank. In other words, for **A** of $n \times n$ dimension, $\mathbf{A}^{-1}$ exists if $\text{rank}(\mathbf{A}) = n$.

Certain Properties of the Rank of a Matrix

- (i) $\text{rank } \mathbf{A} = \mathbf{0} \Leftrightarrow \mathbf{A} = \mathbf{0}$
- (ii) $\text{rank}(\mathbf{I}_n) = n$
- (iii) $\text{rank}(\lambda \mathbf{A}) = \text{rank}(\mathbf{A})$ if $\lambda \neq 0$
- (iv) rank of a diagonal matrix = number of non-zero diagonal elements in the matrix
- (v) $\text{rank}(\mathbf{A} + \mathbf{B}) \leq \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{B})$
- (vi) $\text{rank}(\mathbf{A} - \mathbf{B}) \geq |\text{rank}(\mathbf{A}) - \text{rank}(\mathbf{B})|$
- (vii) $\text{rank}(\mathbf{AB}) \leq \min(\text{rank } \mathbf{A}, \text{rank } \mathbf{B})$

Check Your Progress 1

- 1) Find the inverse of matrix $\mathbf{A} = \begin{bmatrix} 2 & 4 & 1 \\ 4 & 3 & 7 \\ 2 & 1 & 3 \end{bmatrix}$.

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- 2) Find rank of matrix $\mathbf{A} = \begin{bmatrix} 3 & 0 & 2 & 2 \\ -6 & 42 & 24 & 54 \\ 21 & -21 & 0 & -15 \end{bmatrix}$

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3.7 PARTITIONED MATRICES

A partitioned matrix is a matrix of the form

$$Y = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \text{ where } A, B, C, D \text{ are matrices.}$$

All these matrices need not be square however **A** and **B** must have same number of rows. Similarly, **C** and **D** have the same number of rows. The column number of **A** and **C** and **B** and **D** are equal respectively.

Suppose, **A** is $m \times p$ matrix, **B** is $m \times q$ matrix, **C** is $n \times p$ matrix, and **D** is $n \times q$ matrix. The dimension of **Y** is $\{(m+n) \times (p+q)\}$.

A square matrix is 'block-diagonal' if it takes the form where all the blocks are square, but not necessarily of the same order.

For any tow matrices **A** and **B** the direct sum $A \oplus B$ is defined as

$$A \oplus B = \begin{bmatrix} A & O \\ O & B \end{bmatrix} \quad \dots (3.9)$$

In other words, the direct sum of two matrices is a block matrix whose principal diagonals are these two matrices and the off-diagonal matrices are null matrices (need not be of the same order). Thus, a block diagonal matrix is the direct sum of its block. If a square matrix is diagonal then it is block-diagonal but not *vice versa*.

We describe some results pertaining to partitioned matrices.

$$\text{Let } Y_1 = \begin{pmatrix} A_1 & B_1 \\ C_1 & D_1 \end{pmatrix} \quad Y_2 = \begin{pmatrix} A_2 & B_2 \\ C_2 & D_2 \end{pmatrix}$$

Then,

$$Y = Y_1 Y_2 = \begin{pmatrix} A_1 + A_2 & B_1 + B_2 \\ C_1 + C_2 & D_1 + D_2 \end{pmatrix}$$

If the submatrices are well-defined, i.e., Y_1 be an $(m+n) \times (p+q)$ matrix and Y_2 be $(p+q) \times (r+s)$ matrix, then

$$Y = Y_1 Y_2 = \begin{pmatrix} A_1 A_2 + B_1 C_2 & A_1 B_2 + B_1 D_2 \\ C_1 A_2 + D_1 C_2 & C_1 B_2 + D_1 D_2 \end{pmatrix}$$

If $Z = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ then $Z' = \begin{pmatrix} A' & C' \\ B' & D' \end{pmatrix}$ with order of the matrices as follows: $A(m \times p)$, $B(m \times q)$, $C(n \times p)$ and $D(n \times q)$.

Suppose, **A** and **D** are square matrices, but not necessarily of the same order. In that case,

$$\text{tr} \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \text{tr} A + \text{tr} B \quad \dots (3.10)$$

If **A** and **D** are non-singular matrices, then

$$\begin{pmatrix} A & O \\ O & D \end{pmatrix}^{-1} = \begin{pmatrix} A^{-1} & O \\ O & D^{-1} \end{pmatrix}$$

If **B** and **C** are non-singular matrices, then